

The Demonstrations Are a Verifier: Train-Pair Invariants for ARC-AGI-2 Answer Selection

A zero-parameter CPU check, measured on 80,000 attempts from 68 to 73 AI systems, ablated family by family, and added to the NVARC Kaggle pipeline

Abstract

Every ARC task states its rule through a few demonstration pairs. We turn properties that hold on all demonstrations into a verifier: no model, no training. Six families selected on the ARC-AGI-2 training set hold for 98.2% of ARC-AGI-2 evaluation outputs (99.8% on ARC-AGI-1, 95.0% on ConceptARC), flag 24% of wrong attempts from 68 frontier systems and 0.8% of correct ones. As a one-vote penalty it never lowered average pass@2 in pooled selection, even with unsafe families, and raised pass@1 for 24 of 68 systems; ablations show that safety comes from the penalty, not the family list, and that hand-designed invariants see few near misses. Our NVARC-based submission ships it (public score 29.72); on its pools for 109 evaluation tasks it flags 170 candidates, none correct, and changes no score.

1. Introduction

Strong ARC-AGI-2 systems produce many candidates but may submit two. Pooling eight random frontier systems puts a correct answer in the pool for 77% of public evaluation tasks, yet majority vote picks it for only 56%. How much of that gap can the demonstrations close, how reliable are such checks on a benchmark built to defeat shortcuts, and when do they hurt?

2. Prior work

Winning Kaggle systems score candidates with the generator: augmented likelihood (ARChitects 2024, Franzen et al. 2025) or NVARC's kgmon (2025 winner): augmented-decode hits minus augmented NLL. Program synthesis keeps programs that reproduce every train pair (icecuber 2020, Greenblatt 2024, SOAR 2025) but cannot check transductive outputs. The ARC-AGI-2 technical report's pipeline (27%; Lemes de Oliveira et al., arXiv:2603.06590) prunes decoded candidates with colour, size and inclusion filters (about 1% degradation on internal tests). No public work measures how often such invariants transfer to ARC-AGI-2 test outputs, family by family, or their effect on selection.

3. Approach

Invariant families. A family is *inferred* when it holds on every train pair; a candidate is *flagged* when it violates an inferred family. Sixteen families: **shape** (train-consistent size rules over input features), **palette** (exact colour delta or constant palette; subset; keep), **counts** (histogram equal; non-background count equal, non-decreasing, non-increasing), **content** (non-background kept; background kept; crop of input; input embedded in output) and four output **symmetries**.

Safe families. On the ARC-AGI-2 *training* set (1,076 test outputs) we keep families inferred at least 30 times that hold at least 99% of the time: `palette_exact`, `palette_subset`, `hist_equal`, `nonbg_ge`, `keep_nonbg`, `input_in_output`. Shape (98.6%), `keep_bg` (95.1%) and `nonbg_equal` (96.9%) miss the bar; the set was fixed before any evaluation-set measurement.

Using flags. (a) *Attempt order*: swap the two attempts when only attempt 1 is flagged. (b) *Pool selection*: rank candidates by support (votes, or NVARC hits) minus $\lambda \cdot \text{flag}$; $\lambda = \infty$ is a hard veto; we use $\lambda = 1$, one vote per violated invariant.

Kaggle submission. We fork the NVARC 2025 notebook (Qwen3-4B, per-task test-time training, augmented decoding, kgmon; appendix) and change only the final ranking, to kgmon – 1·flag; with no flags the output is identical.

4. Why it works, and when it hurts

Soundness. A task applies one rule to every pair, so a property implied by the rule also holds on the test. A property holding on k demonstrations *by coincidence* must get lucky k times; per-pair coincidence rates (pass rates on unrelated outputs) are 0.4% for `palette_exact` and 0% for `keep_nonbg` and `hist_equal`. The remaining failures are **coverage gaps**: the demonstrations never show the part of the rule the test needs, as in all three ARC-AGI-2 evaluation failures (446ef5d2: the test removes a colour no demonstration removed). Gaps shrink with more demonstrations (failures 1.2% at $k=2$, 0% at $k \geq 5$ on ARC-AGI-2 training; 7.6% to 3% on ConceptARC), and so does recall, since each pair can only remove families (31% at $k=2$, 13% at $k=4$).

Selection risk. Let $\varepsilon = P(\text{flag} \mid \text{correct})$. Raising λ changes a task only when the correct answer c^* crosses the top-2 boundary: a **gain** needs a flagged wrong candidate less than λ above an unflagged c^* ; a **loss** needs a flagged c^* with two unflagged candidates less than λ below it. A hard veto's loss is about $\varepsilon \cdot P(c^* \text{ already in top-2})$, growing as the base selector improves while the gain shrinks: vetoes help weak pools and hurt saturated ones. With $\lambda=1$ a loss also needs a tie on support, making it second-order.

5. Results

Intervals: 95% task bootstraps; tables in the appendix.

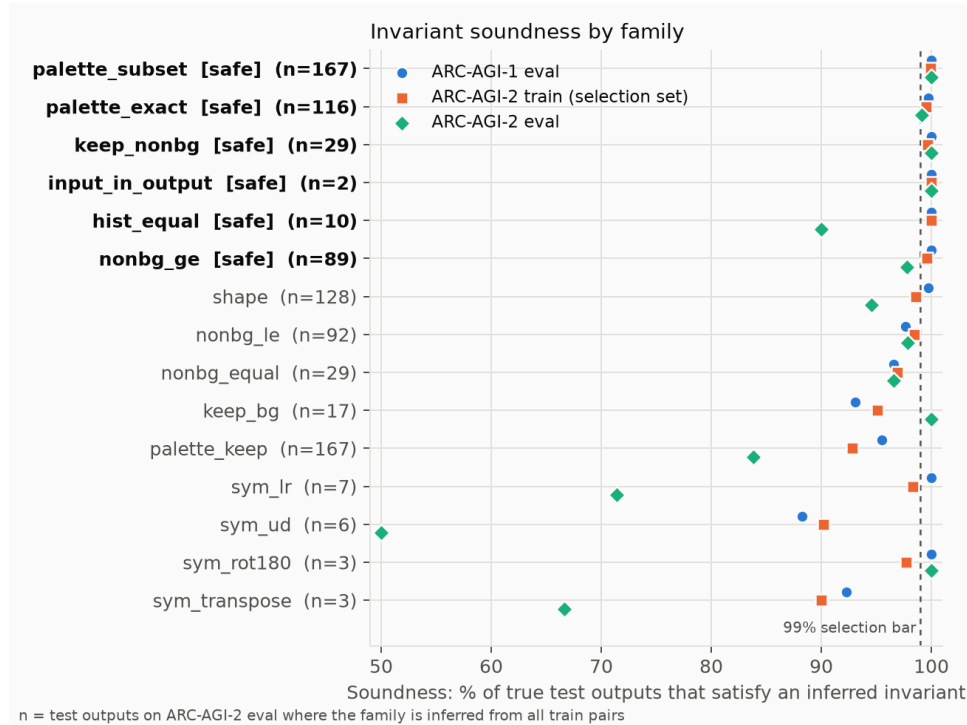


Fig. 1

Soundness (Fig. 1). The six safe families jointly hold for 99.2% of ARC-AGI-2 training, 98.2% of ARC-AGI-2 evaluation, 99.8% of ARC-AGI-1 evaluation and 95.0% of ConceptARC outputs (160 tasks by other authors), and reject 91% of unrelated outputs.

Which attempts the verifier flags

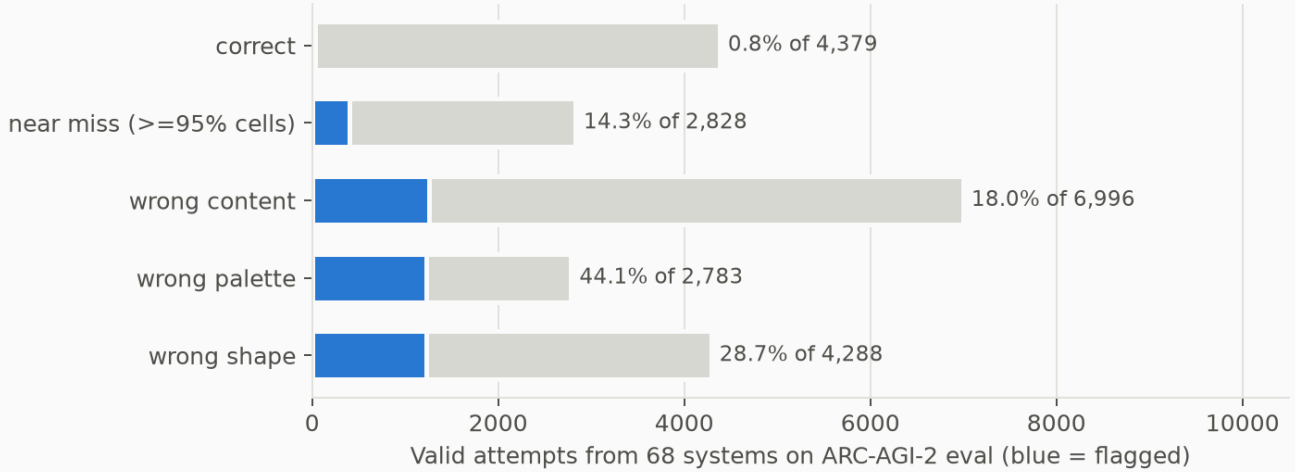


Fig. 2

Error taxonomy (Fig. 2). Of 22,320 ARC-AGI-2 attempt slots from 68 systems (classes in Fig. 2; near miss: $\geq 95\%$ of cells right), the verifier flags 24.4% of valid wrong attempts (44% of wrong-palette, 14% of near misses) and 0.8% of correct ones, all on the three coverage-gap outputs; on ARC-AGI-1 (57,854 slots, 73 systems) 23.9% and 0.4%.

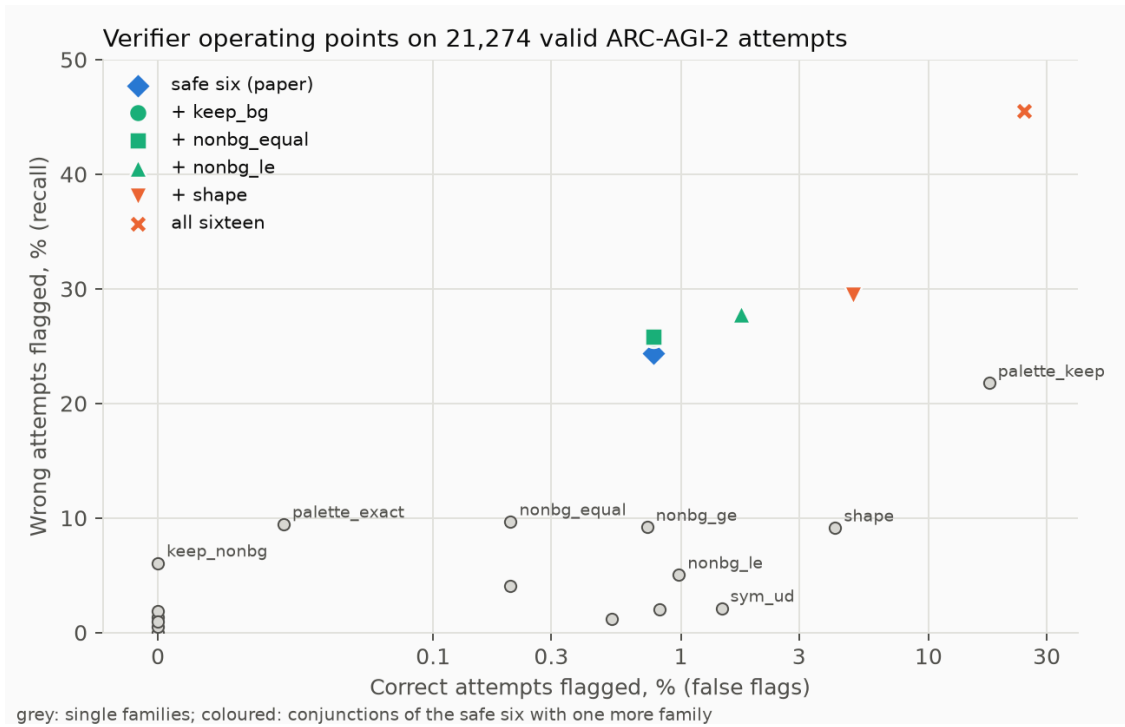


Fig. 4

Which families matter (Fig. 4). Removing palette_exact costs the most recall (24.4% to 17.6%), then keep_nonbg (19.1%) and nonbg_ge (19.5%). All sixteen families reach 45.5% recall but flag 24.3% of correct answers and see only 27% of near misses. keep_bg and nonbg_equal, excluded by the training bar, would have added recall for free (25.7%, 25.8%).

Attempt order. Swapping flagged first attempts: ARC-AGI-2, 24 of 68 systems improved, none got worse, +0.51 pass@1 points [0.36, 0.65]; ARC-AGI-1, 61 of 73 improved, +1.12 [0.98, 1.28].

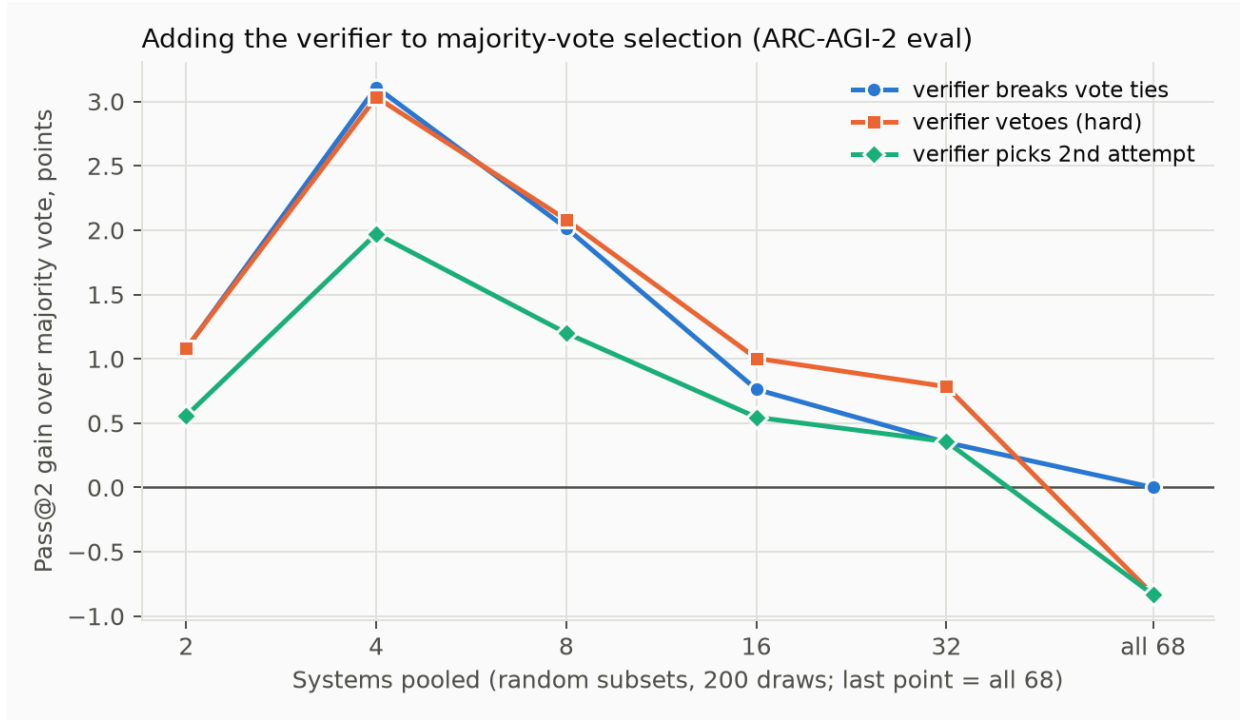


Fig. 3

Pooled selection (Fig. 3). Pass@2 (%) on ARC-AGI-2 for random pools, 200 draws per size:

Systems	Vote	One-vote penalty	Hard veto
2	36.4	+1.1 [0.7, 1.4]	+1.1
4	42.9	+3.1 [2.4, 3.9]	+3.0
8	56.3	+2.0 [1.5, 2.6]	+2.1
16	72.5	+0.8 [0.4, 1.1]	+1.0
32	80.7	+0.4 [0.1, 0.7]	+0.8
all 68	84.9	0.0	-0.8 [-2.5, 0.0]

The one-vote penalty is never negative on average, losing to plain voting in 14 of 1,000 random pools; the hard veto turns negative on the full pool, on the coverage-gap tasks. On ARC-AGI-1 (coverage gaps 0.24%) both rules are non-negative at every size.

Ablations. Removing `palette_exact` costs 1.0 of the 3.7 points gained at 4 systems; adding `keep_bg`, `nonbg_equal` or `nonbg_le` adds 0.3 to 0.4. Under the one-vote penalty even all sixteen families stay non-negative at every size (+3.3 at 4, +0.6 at 16, 0.0 at 68); as a hard veto they lose 7 points at 16 systems and 14 at 68; ARC-AGI-1 repeats the pattern (appendix). Pooling the strongest systems by pass@2 (top 4 vote at 91.9, above 84.9 for all 68), the penalty adds +0.2 (top 4), +1.1 (top 8), +0.3 (top 16); the hard veto loses on the top 4 and 16.

Theory check. Per output, hard-veto losses rise with pool size from 0.09% (2 systems) to 0.60% (all 68), below ϵ -pass@2, while gains fall from 3.8% to 0.06%; one-vote losses vanish. The penalty closes 17 to 20% of the vote-to-oracle gap at 2 to 4 systems and under 5% beyond 16.

NVARC's own candidates. A 5-hour copy of the submission notebook covered 109 of the 120 evaluation tasks: 146 outputs, 1,101 distinct candidates, 53 correct. It flags 170, all wrong (66 wrong-palette, 59 near misses), none correct; it reorders two outputs' top 2 and holds the score at 31.7/109 for $\lambda \leq 2$. A hard veto reorders 17 and gains one task, as expected for an unsaturated pool.

Kaggle leaderboard. Public score **29.72**, submission **56275620**, notebook version 2; unmodified NVARC copies score 30.6 to 31.8.

6. Limitations

- **Pool source:** frontier-LLM attempts; NVARC pools cover 109 of 120 tasks.
- **Penalty:** $\lambda=1$ suits integer votes; kgmon mixes hits with likelihood and may want calibration.
- **Recall:** 24% of wrong LLM answers; sixteen families see only 27% of near misses.
- **Sample size:** 120 evaluation tasks; intervals cover task sampling only.

7. Conclusion

The demonstrations already contain a free verifier, sound on 98% of ARC-AGI-2 evaluation outputs and strong enough to lift pass@1 and small-pool selection. What makes it safe is the penalty, not the family list: one vote per violated invariant stays non-negative even with unsafe families; a veto does not. Its blind spot is the near miss, which needs a verifier that sees structure the demonstrations only imply.

AI assistance and reproducibility

AI-assisted (Claude Code), human-directed. All code and this text were written by Claude (Anthropic) from the author's prompts, which set the question, families, safety criterion, experiments and Kaggle constraints; the author reviewed every step and is responsible for all claims. Analysis is CPU-only. Data: arcprize/ARC-AGI-2 (Apache-2.0), fchollet/ARC-AGI, ConceptARC (Moskvichev et al. 2023), arcprize attempt datasets (MIT). Code: the notebook in Project Links; analysis, results, appendix and this text at <https://github.com/CVasilopoulos/arc-agi2-invariant-verifier> (CC BY 4.0).